

## URUMQI, China

WMO: 514630

Lat: 43.783N

Lon: 87.617E

Elev: 919

StdP: 90.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -22.1      | -19.8 | -25.1                         | 0.4 | -21.2 | -22.6 | 0.5 | -19.2 | 5.4                   | -8.2 | 4.6 | -9.1 | 1.5                   | 90   | 0.482 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.0                    | 33.7            | 16.7 | 32.1 | 16.3 | 30.7 | 15.9 | 18.2                | 28.4 | 17.6 | 27.9 | 17.0 | 27.3 | 3.3                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 15.3                            | 12.1 | 20.7 | 14.2 | 11.3 | 20.8 | 13.2 | 10.6 | 21.2 | 55.3          | 28.4 | 53.1 | 27.8 | 51.3 | 27.3 | 21.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.4           | 5.8  | 4.8 | DB                         | -24.0 | 37.0               | 2.4 | 1.7                                                | -25.7 | 38.2       | -27.1 | 39.2       | -28.5 | 40.2       | -30.3 | 41.4 | (4) |
| (5)               |      |     | WB                         | -24.2 | 19.4               | 2.4 | 1.1                                                | -25.9 | 20.2       | -27.3 | 20.9       | -28.7 | 21.5       | -30.4 | 22.3 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.1   | -12.2 | -8.8  | 0.8   | 11.5  | 17.4  | 22.5  | 24.4  | 23.1  | 17.6  | 9.3   | -0.2  | -8.9 | (6)  |
| (7)  |                                                                                  | DBStd       | 13.74 | 4.70  | 4.59  | 6.38  | 5.39  | 4.55  | 3.57  | 3.21  | 3.51  | 4.60  | 5.07  | 5.53  | 4.72 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2527  | 687   | 527   | 293   | 47    | 4     | 0     | 0     | 0     | 4     | 75    | 307   | 584  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4291  | 945   | 761   | 545   | 209   | 71    | 8     | 2     | 5     | 66    | 281   | 556   | 843  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1852  | 0     | 0     | 6     | 92    | 234   | 377   | 447   | 407   | 233   | 54    | 1     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 573   | 0     | 0     | 0     | 4     | 43    | 134   | 191   | 154   | 45    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 5182  | 0     | 0     | 0     | 42    | 343   | 1192  | 1843  | 1418  | 336   | 7     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 1854  | 0     | 0     | 0     | 7     | 79    | 408   | 733   | 539   | 87    | 1     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.3   | 1.7   | 1.8   | 2.2   | 2.8   | 2.8   | 2.7   | 2.6   | 2.5   | 2.6   | 2.2   | 1.9   | 1.7  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 267   | 9     | 10    | 19    | 34    | 36    | 32    | 29    | 22    | 22    | 20    | 15    |      | (15) |
| (16) |                                                                                  | PrecMax     | 420   | 26    | 32    | 47    | 90    | 126   | 95    | 108   | 87    | 66    | 67    | 45    | 51   | (16) |
| (17) |                                                                                  | PrecMin     | 131   | 0     | 0     | 1     | 0     | 0     | 2     | 0     | 1     | 0     | 0     | 4     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 77    | 6     | 6     | 11    | 21    | 25    | 22    | 23    | 20    | 16    | 14    | 11    | 11   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 0.8   | 3.8   | 19.1  | 27.1  | 31.2  | 34.1  | 36.7  | 35.5  | 32.2  | 24.1  | 15.3  | 3.9  | (19) |
| (20) |                                                                                  |             | MCWB  | -1.8  | -0.6  | 8.1   | 11.7  | 14.2  | 16.2  | 17.2  | 16.6  | 15.3  | 11.0  | 7.3   | -0.4 | (20) |
| (21) |                                                                                  | 2%          | DB    | -1.8  | 0.6   | 15.7  | 24.2  | 28.8  | 32.2  | 34.0  | 33.3  | 29.1  | 21.0  | 11.5  | 0.8  | (21) |
| (22) |                                                                                  |             | MCWB  | -3.8  | -1.8  | 7.4   | 10.7  | 13.5  | 15.8  | 17.2  | 16.4  | 14.2  | 10.2  | 5.3   | -1.6 | (22) |
| (23) |                                                                                  | 5%          | DB    | -3.8  | -1.3  | 12.8  | 21.8  | 27.0  | 30.7  | 32.2  | 31.6  | 26.8  | 19.0  | 9.1   | -1.2 | (23) |
| (24) |                                                                                  |             | MCWB  | -5.3  | -3.1  | 6.3   | 9.9   | 13.1  | 15.7  | 16.8  | 16.2  | 13.4  | 9.3   | 4.0   | -3.0 | (24) |
| (25) |                                                                                  | 10%         | DB    | -5.9  | -2.9  | 10.1  | 19.7  | 25.1  | 29.1  | 30.6  | 29.8  | 25.0  | 16.8  | 6.9   | -2.8 | (25) |
| (26) |                                                                                  |             | MCWB  | -7.2  | -4.4  | 5.1   | 9.0   | 12.4  | 15.3  | 16.5  | 15.9  | 12.9  | 8.6   | 2.9   | -4.3 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | -1.4  | 0.2   | 9.3   | 13.0  | 15.6  | 18.2  | 20.0  | 18.7  | 16.4  | 12.1  | 7.6   | 0.3  | (27) |
| (28) |                                                                                  |             | MCDWB | 0.2   | 2.6   | 16.9  | 23.4  | 26.9  | 28.1  | 31.1  | 27.4  | 27.8  | 21.0  | 13.9  | 2.5  | (28) |
| (29) |                                                                                  | 2%          | WB    | -3.6  | -1.7  | 7.8   | 11.6  | 14.7  | 17.3  | 18.6  | 17.8  | 15.2  | 10.9  | 5.7   | -1.4 | (29) |
| (30) |                                                                                  |             | MCDWB | -2.0  | 0.3   | 14.5  | 20.7  | 25.6  | 27.6  | 28.9  | 28.4  | 26.4  | 19.2  | 10.9  | 0.5  | (30) |
| (31) |                                                                                  | 5%          | WB    | -5.3  | -3.0  | 6.6   | 10.7  | 14.0  | 16.7  | 17.9  | 17.2  | 14.2  | 9.9   | 4.3   | -2.9 | (31) |
| (32) |                                                                                  |             | MCDWB | -3.8  | -1.2  | 12.1  | 19.4  | 24.3  | 27.2  | 28.2  | 27.8  | 25.2  | 17.9  | 8.5   | -1.4 | (32) |
| (33) |                                                                                  | 10%         | WB    | -7.1  | -4.4  | 5.3   | 9.8   | 13.2  | 16.1  | 17.3  | 16.6  | 13.4  | 8.8   | 3.1   | -4.2 | (33) |
| (34) |                                                                                  |             | MCDWB | -6.0  | -3.0  | 9.9   | 18.4  | 23.1  | 26.4  | 27.4  | 27.2  | 23.7  | 16.0  | 6.7   | -3.0 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.7   | 6.5   | 7.1   | 9.2   | 9.3   | 9.0   | 9.0   | 9.4   | 9.4   | 8.2   | 6.0   | 6.3  | (35) |
| (36) |                                                                                  |             | MCDBR | 9.1   | 8.6   | 10.3  | 11.9  | 11.0  | 10.9  | 11.0  | 11.6  | 11.3  | 10.5  | 8.9   | 9.0  | (36) |
| (37) |                                                                                  |             | MCWBR | 7.3   | 6.2   | 5.4   | 4.9   | 4.1   | 3.5   | 3.4   | 3.6   | 4.2   | 4.7   | 5.1   | 6.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWR | 8.9   | 8.2   | 9.5   | 10.4  | 9.7   | 8.9   | 9.0   | 9.4   | 10.2  | 9.8   | 8.4   | 8.3  | (38) |
| (39) |                                                                                  |             | MCWBR | 7.1   | 6.0   | 5.2   | 4.5   | 3.9   | 3.0   | 3.2   | 3.3   | 3.9   | 4.5   | 4.9   | 6.2  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.333 | 0.386 | 0.444 | 0.498 | 0.488 | 0.498 | 0.479 | 0.460 | 0.435 | 0.416 | 0.371 | 0.333 | (40) |      |
| (41) |                                                                                  | taud        | 2.088 | 1.989 | 1.919 | 1.846 | 1.921 | 1.959 | 2.052 | 2.087 | 2.077 | 2.048 | 2.102 | 2.127 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 754   | 771   | 776   | 768   | 792   | 785   | 795   | 796   | 780   | 729   | 700   | 712   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 103   | 137   | 167   | 195   | 187   | 180   | 163   | 152   | 141   | 127   | 102   | 91    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.01  | 2.92  | 4.23  | 5.50  | 6.40  | 6.53  | 6.38  | 5.87  | 5.02  | 3.53  | 2.19  | 1.64  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.09  | 0.17  | 0.26  | 0.35  | 0.32  | 0.34  | 0.30  | 0.27  | 0.17  | 0.19  | 0.18  | 0.13  | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (4 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | -0.69 | N/A         | N/A     | +131    | +90     |

Nomenclature: See separate page